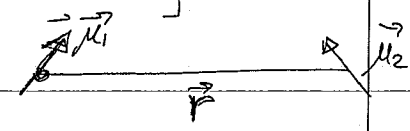
Magnetic interaction

① Dipolar $E = \frac{\mu_0}{4\pi r^3}\left[\vec{\mu}_1 \cdot \vec{\mu}_2 - \frac{3}{r^2}(\vec{\mu}_1 \cdot \vec{r})(\vec{\mu}_2 \cdot \vec{r})\right]$



For $|\vec{\mu}| = \mu_B$ the order of magnitude is estimated as $E \simeq \frac{\mu_0}{4\pi r_0^3}\mu_B^2$ where we can take $r_0 = 1\,\text{Å} = 10^{-10}\,m$.

Now, $\mu_0 = 4\pi \cdot 10^{-7}$ Henry/m $= 4\pi \cdot 10^{-7} \frac{kg \cdot m}{s^2 (\text{Ampere})^2}$

$$\mu_B = 9.274 \cdot 10^{-24}\ \text{Ampere} \cdot m^2 \simeq 10^{-23}\ A\,m^2.$$

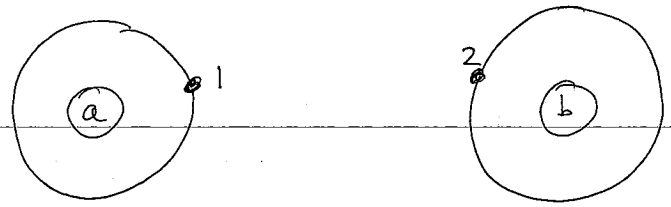
$$\Rightarrow E \simeq 10^{-7} \frac{kg \cdot m}{s^2 A^2} \frac{(10^{-23}\ A \cdot m^2)^2}{(10^{-10}\ m)^3} = 10^{-23} \frac{kg \cdot m^2}{s^2} = 10^{-23}\,J$$

but Joule $= 7.24 \cdot 10^{22}\ K \Rightarrow 10^{-23}\,J \simeq 1\,K.$

Thus dipolar energy $\lesssim 1\,K$ for magnetic moments one atomic size (1 Å) apart. Note also that it decays fast with distance: at 2 Å the energy will be 8 times smaller, almost an order of magnitude!

Most magnetic materials have ordering temperatures of the order of hundreds and even thousands of Kelvins $\Rightarrow$ magnetic dipolar interaction is <u>too</u> small to explain this.

Exchange interaction



Two atoms/ions/ quantum wells : a and b.

Wave function of electron in a : $\varphi_a(r)$.

For two electrons : $\varphi_a(r_1)\,\varphi_b(r_2)$

But this assumes distinguishable particles:

electron #1 is in state φ_a while electron #2 → in φ_b.

Electrons are not distinguishable, and their total wave function (ψ(spatial) $\cdot$ χ(spin)) must be antisymmetric under exchange.

From addition of two spin angular momenta we know that χ(spin) can only be χ_S = singlet (non-degenerate)

OR

χ_T = triplet (3-fold degenerate)

$$\chi_S = \frac{1}{\sqrt{2}}\left|\uparrow_1\downarrow_2 - \downarrow_1\uparrow_2\right\rangle \longrightarrow \text{antisymmetric}$$

$$\chi_T = \left\{\begin{array}{l} |\uparrow_1\uparrow_2\rangle \\ \frac{1}{\sqrt{2}}\left|\uparrow_1\downarrow_2 + \downarrow_1\uparrow_2\right\rangle \\ |\downarrow_1\downarrow_2\rangle \end{array}\right\} \text{symmetric under } 1 \leftrightarrow 2 \text{ exchange.}$$

$$\chi_T(1,2) = \chi_T(2,1)$$

$$\chi_S(1,2) = -\chi_S(2,1)$$

Thus χ_S must multiply <u>symmetric spatial</u> w.f.

$$\psi_S = \frac{1}{\sqrt{2}}\left(\psi_a(r_1)\,\psi_b(r_2) + \psi_a(r_2)\,\psi_b(r_1)\right)$$

while χ_T must multiply <u>antisymmetric spatial</u> w.f.

$$\psi_T = \frac{1}{\sqrt{2}}\left(\psi_a(r_1)\,\psi_b(r_2) - \psi_a(r_2)\,\psi_b(r_1)\right)$$

this is assuming $\langle \psi_a | \psi_b \rangle = 0$

Observe that $\psi_a(r_2)\,\psi_b(r_1) = \hat{P}_{12}\,\psi_a(r_1)\,\psi_b(r_2)$

exchange operator (exchanges positions of particles 1 and 2)

$\boxed{\text{Heitler–London approximation}}$

let now $H = H_1 + H_2 + H'$

$$H_1 = -\frac{\hbar^2 \nabla_1^2}{2m} - \frac{Ze^2}{4\pi\varepsilon_0 |\vec{r}_1 - \vec{r}_a|}$$

location of atom a.

coordinate of electron 1

$$H_2 = -\frac{\hbar^2 \nabla_2^2}{2m} - \frac{Ze^2}{4\pi\varepsilon_0 |\vec{r}_2 - \vec{r}_b|}$$

$$H' = \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_1 - \vec{r}_2|} - \frac{Ze^2}{4\pi\varepsilon_0 |\vec{r}_1 - \vec{r}_b|} - \frac{Ze^2}{4\pi\varepsilon_0 |\vec{r}_2 - \vec{r}_a|}$$

repulsion between electrons

attraction of 1 to atom b

attraction of 2 to atom a

Proper normalization of $\psi_{S,T}$

$$\langle \psi_S | \psi_S \rangle = 1 \Rightarrow \frac{|C_S|^2}{2} \langle \psi_a(1)\psi_b(2) + \psi_a(2)\psi_b(1) | \psi_a(1)\psi_b(2) + \psi_a(2)\psi_b(1) \rangle$$

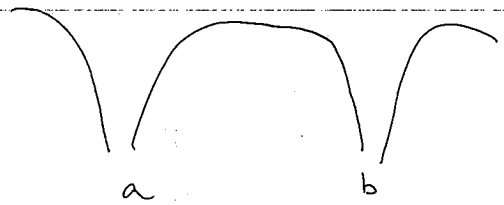
$$= \frac{|C_S|^2}{2} \Big\{ \langle \psi_a(1)|\psi_a(1)\rangle \langle \psi_b(2)|\psi_b(2)\rangle + \langle \psi_a(2)|\psi_a(2)\rangle \langle \psi_b(1)|\psi_b(1)\rangle$$

$$+ \langle \psi_a(1)|\psi_b(1)\rangle \langle \psi_b(2)|\psi_a(2)\rangle + \langle \psi_a(2)|\psi_a(1)\rangle \langle \psi_b(1)|\psi_b(2)\rangle \Big\}$$

$$= \frac{|C_S|^2}{2}\left(1 + 1 + S S^* + S^* S\right) = \left(1 + |S|^2\right) |C_S|^2 \Rightarrow C_S = \frac{1}{\sqrt{1+S^2}}$$

where $S = \langle \psi_a(1) | \psi_b(1) \rangle$ = overlap integral

$$= \int d\vec{r}\, \psi_a^*(\vec{r})\, \psi_b(\vec{r})$$



Similarly, $\langle \psi_T | \psi_T \rangle = 1 \Rightarrow C_T = \dfrac{1}{\sqrt{1-S^2}}$

$$\psi_{S,T} = \frac{1}{\sqrt{2}\sqrt{1 \pm S^2}} \left(\psi_a(r_1)\psi_b(r_2) \pm \psi_a(r_2)\psi_b(r_1) \right).$$

Note that by construction $H_1 \psi_a(r_1) = E_a \psi_a(r_1)$

$$H_2 \psi_b(r_2) = E_b \psi_b(r_2)$$

i.e. $\psi_{a,b}$ are eigenstates of single-atom (single-center) problem.

Energy of singlet:

$$E_S = \langle \psi_S | H | \psi_S \rangle = \frac{1}{2(1+S^2)} \langle a(1)b(2) + a(2)b(1) | \left(H_1 + H_2 + H'\right) \cdot$$

$$\cdot | a(1)b(2) + a(2)b(1) \rangle$$

Consider, for example, term $\langle a(1)\, b(2) \,|H|\, a(2)\, b(1)\rangle =$

$$= \int \psi_a(r_1)\, \psi_b(r_2) \left(\underbrace{-\frac{\hbar^2 \nabla_1^2}{2m} - \frac{Ze^2}{|r_1 - r_a|}}_{H_b} \underbrace{- \frac{Ze^2}{|r_1 - r_b|} - \frac{\hbar^2 \nabla_2^2}{2m} - \frac{Ze^2}{|r_2 - r_b|}}_{H_a} - \frac{Ze^2}{|r_2 - r_a|} + \frac{e^2}{|r_1 - r_2|} \right) \psi_a(r_2)\, \psi_b(r_1) \Rightarrow$$

But $\left(-\frac{\hbar^2 \nabla_1^2}{2m} - \frac{Ze^2}{(r_1 - r_b)} \right) \psi_b(r_1) = E_b\, \psi_b(r_1)$

$$\Rightarrow \int \psi_a(r_1)\, \psi_b(r_2) \left(E_a + E_b + H' \right) \psi_a(r_2)\, \psi_b(r_1) =$$

$$= (E_a + E_b) \underbrace{\int \psi_a(r_1)\, \psi_b(r_1)\, dr_1}_{S} \underbrace{\int \psi_b(r_2)\, \psi_a(r_2)\, dr_2}_{S^*} + \langle a(1)\, b(2) | H' | a(2)\, b(1) \rangle$$

$$\Rightarrow E_s = E_a + E_b + \left[\langle a(1)\, b(2) | H' | a(2)\, b(1) \rangle + \langle a(1)\, b(2) | H' | a(1)\, b(2) \rangle \right] \frac{1}{1+s^2}.$$

Similarly, $E_t = E_a + E_b + \left[\langle a(1)\, b(2) | H' | a(1)\, b(2) \rangle - \langle a(1)\, b(2) | H' | a(2)\, b(1) \rangle \right] \frac{1}{1-s^2}$

Denote $K \equiv \langle \psi_a(1)\, \psi_b(2) | H' | \psi_a(1)\, \psi_b(2) \rangle$ (Coulomb integral)

$J \equiv \langle \psi_a(1)\, \psi_b(2) | H' | \psi_a(2)\, \psi_b(1) \rangle$ (exchange integral)

$$E_s - E_t = \frac{K + J}{1+s^2} - \frac{K - J}{1-s^2} = \frac{-Ks^2 + J}{1-s^4} \cdot (2)$$

Although it may not seem obvious, $E_s < E_t$ <u>always</u> in a two-electron problem: $E_s - E_t < 0$.

[This is Theorem: the ground state of $H = -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) + V(\vec{r}_1, \vec{r}_2)$ is <u>nodeless</u> [ignoring statistics, or for distinguishable particles]

$\Downarrow$

spatial part is <u>symmetric</u> $\Rightarrow$ singlet!

Let's check this for a simplified case of no overlap $\langle a|b\rangle = 0$, when

$$S = \int dr\, \psi_a(r)\, \psi_b(r) = 0.$$

$$\Rightarrow E_s - E_t = 2J = 2\int dr_1 dr_2\, \psi_a^*(r_1)\, \psi_b^*(r_2) \left(\frac{e^2}{4\pi\varepsilon_0}\right)$$

$$\left(\frac{1}{|r_1-r_2|} - \frac{Z}{|r_1-r_b|} - \frac{Z}{|r_2-r_a|}\right) \psi_a(r_2)\, \psi_b(r_1) =$$

$$= 2\,\frac{e^2}{4\pi\varepsilon_0} \int dr_1 dr_2\, \psi_a^*(r_1)\, \psi_b^*(r_2)\, \frac{1}{|\vec{r}_1-\vec{r}_2|}\, \psi_b(r_1)\, \psi_a(r_2)$$

$$= 2\,\frac{e^2}{4\pi\varepsilon_0} \int dr_1 dr_2\, \psi_a^*(r_1)\, \psi_b(r_1)\, \frac{1}{|\vec{r}_1-\vec{r}_2|}\, \psi_b^*(r_2)\, \psi_a(r_2)$$

Denote $f_{ab}(r) = \psi_a^*(r)\, \psi_b(r)$

Then $J = \frac{e^2}{4\pi\varepsilon_0} \int dr_1 dr_2\, f_{ab}(r_1)\, \frac{1}{|\vec{r}_1-\vec{r}_2|}\, f_{ab}^*(r_2) > 0$

no matter what precise form of $f_{ab}(r)$ is.

Thus $E_s - E_t = 2J > 0$: singlet has lower energy.

- Note that J is determined by electrostatic repulsion between two electrons $\Rightarrow$ can be very large!
- On the other hand it clearly is sensitive to the overlap between wave functions centered at $\vec{r}_a$ and $\vec{r}_b$. In fact, it is exponentially sensitive for $|\vec{r}_a - \vec{r}_b| \gg 1$.
- At large distances Heitler-London approximation fails. In a sense that it starts to predict that triplet becomes the ground state, which contradicts to exact theorem. The problem is fixed by C. Herring Rev. Mod. Phys. 34 631 (1962) and L. P. Gorkov and I. Dzyaloshinskii Sov. Phys.-Doklady 8 788 (1964)

Heisenberg Hamiltonian

We see that the energy difference between singlet and triplet states is enforced by their spin structures (which translates into different spatial structure of $\psi_{S,T}$)

On the other hand, singlet and triplet spin wave functions are eigenstates of total spin angular momentum

$$\hat{S}^2_{tot} \frac{1}{\sqrt{2}}|\uparrow\downarrow - \downarrow\uparrow\rangle = \hbar^2 S_{tot}(S_{tot}+1)\frac{1}{\sqrt{2}}|\uparrow\downarrow - \downarrow\uparrow\rangle = 0$$

because $S_{TOT} = 0$ for singlet.

$$\hat{S}^2_{tot}\begin{cases} |\uparrow\uparrow\rangle \\ \frac{1}{\sqrt{2}}|\uparrow\downarrow + \downarrow\uparrow\rangle \\ |\downarrow\downarrow\rangle \end{cases} = \hbar^2 \underset{\substack{\parallel \\ 1}}{S_{TOT}}(S_{TOT}+1)\,|triplet\rangle = 2\hbar^2|triplet\rangle$$

but $$\hat{S}^2_{TOT} = (\vec{S}_1 + \vec{S}_2)^2 = \underbrace{\underbrace{\vec{S}_1^2}_{\frac{3\hbar^2}{4}} + \underbrace{\vec{S}_2^2}_{\frac{3\hbar^2}{4}}} + 2\vec{S}_1\cdot\vec{S}_2$$

~~for states by~~ spin of electron is fixed!

$\Rightarrow$ singlet and triplet are also eigenstates of operator $\vec{S}_1 \cdot \vec{S}_2$:

$$\vec{S}_1\cdot\vec{S}_2 \frac{1}{\sqrt{2}}|\uparrow\downarrow - \downarrow\uparrow\rangle = \frac{1}{2}\Big(\overset{\hat{S}^2_{TOT}}{0} - 2\cdot\frac{3\hbar^2}{4}\Big)\frac{1}{\sqrt{2}}|\uparrow\downarrow - \downarrow\uparrow\rangle =$$

$$= -\frac{3\hbar^2}{4}\frac{1}{\sqrt{2}}|\uparrow\downarrow - \downarrow\uparrow\rangle$$

$$\vec{S}_1\cdot\vec{S}_2 \begin{cases} \uparrow\uparrow \\ \uparrow\downarrow + \downarrow\uparrow \\ \downarrow\downarrow \end{cases} = \frac{1}{2}\Big(2\hbar^2 - 2\,\frac{3\hbar^2}{4}\Big)|triplet\rangle =$$

$$= \frac{\hbar^2}{4}|triplet\rangle$$

Note that the energy difference is $1\hbar^2$.

We can thus write Hamiltonian for spin degrees of freedom:

$$\left.\begin{array}{l} |\text{singlet}\rangle \text{ has energy } E_s. \\ |\text{triplet}\rangle \text{ has energy } E_t. \end{array}\right\} \text{ their difference } E_s - E_t$$

$\Rightarrow$ for any two spins the spin Hamiltonian is

$$H_{Heis} = J\, \vec{S}_1 \cdot \vec{S}_2$$

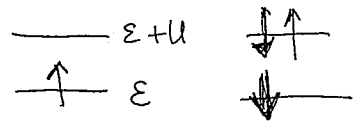
where $\frac{J}{\hbar^2} = E_t - E_s > 0$.

Acting on spin states, $\frac{J}{\hbar^2} \vec{S}_1 \vec{S}_2 |\chi\rangle = \begin{cases} -\frac{3}{4} J |\chi\rangle \text{ for } \chi = \text{singlet} \\ \frac{1}{4} J |\chi\rangle \text{ for } \chi = \text{triplet} \end{cases}$

Different mutual orientation of spins leads to different electrostatic energies (due to antisymmetry of the full ~~wa~~ ~~f~~ electron wavefunction) $\Rightarrow$ $\boxed{\text{this is the origin of magnetic properties of } \underline{\text{solids}}}$

Superexchange via Hubbard model.

Hubbard model : 1 orbital per site (2 spin states)



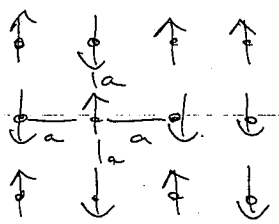
energy of one electron $= \varepsilon$

energy of two electrons $= \varepsilon + U$

$$H_0 = \varepsilon\,|\uparrow\rangle\langle\uparrow| + \varepsilon\,|\downarrow\rangle\langle\downarrow| + U\,|\uparrow\downarrow\rangle\langle\uparrow\downarrow|$$

Note that states like $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ are forbidden by Pauli principle.

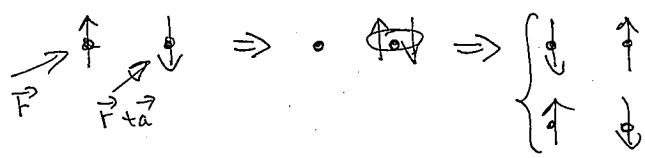
<u>Zeroth approximation</u>: every site has 1 electron, with spin $\uparrow$ or $\downarrow$

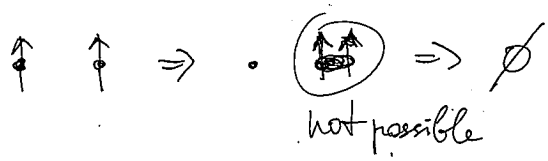


$\Rightarrow 2^N$ fold degenerate manifold

Now we allow for <u>hopping</u> of electrons : $H_t = -t\left(|\vec{r}\rangle\langle\vec{r}+\vec{a}| + |\vec{r}+\vec{a}\rangle\langle\vec{r}|\right)$

Consider just two sites





Thus <u>antiparallel</u> electrons on nearest sites gain energy by <u>virtual processes</u> when the energy of intermediate state is $\varepsilon + U$.

$$\Delta E = \sum \frac{|\langle n|V|0\rangle|^2}{E_0 - E_n}$$

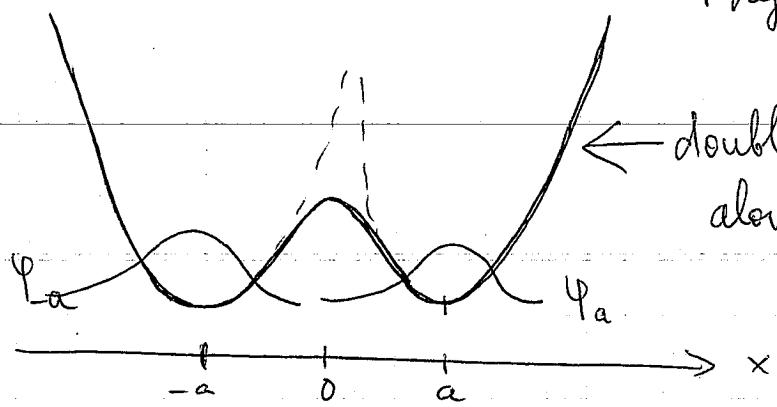
here $V = H_t$: ground state $|0\rangle = |\uparrow\vec{r}+\vec{a}\rangle|\downarrow\vec{r}\rangle$ or $|\downarrow\vec{r}+\vec{a}\rangle|\uparrow\vec{r}\rangle$

excited state $|n\rangle = |0\,\vec{r}+\vec{a}\rangle|\uparrow\downarrow\,\vec{r}\rangle$ or $|\uparrow\downarrow\vec{r}+\vec{a}\rangle|0\,\vec{r}\rangle$

$\langle 0|H_t|n\rangle = -t$; $E_n = \varepsilon + U$

$\Rightarrow \Delta E = -\dfrac{t^2}{U}$ <u>energy gain!</u> This also relieves degeneracy.

Semiconducting quantum dots: Burkard, Loss, DiVincenzo
Phys Rev B 59 2070 (1999)



double well potential along x; simple oscillator along y.

Confining potential $V(x,y) = \frac{1}{2} m\omega^2 \left(\frac{(x^2-a^2)^2}{4a^2} + y^2 \right)$

$$\psi_\pm(r_1, r_2) = \frac{1}{\sqrt{2(1 \pm S^2)}} \left[\varphi_a(r_1)\varphi_{-a}(r_2) \pm \varphi_a(r_2)\varphi_a(r_1) \right]$$

Two-particle Hamiltonian:

$$H_{12} = H_{01} + H_{02} + V_{12}$$

$$H_{01} = \frac{\vec{p}_1^{\,2}}{2m} + \frac{1}{2} m\omega^2 \left((x_1 - a)^2 + y_1^2 \right)$$

$$H_{02} = \frac{\vec{p}_2^{\,2}}{2m} + \frac{1}{2} m\omega^2 \left((x_2 + a)^2 + y_2^2 \right)$$

<u>Perturbation</u> $V_{12} = \frac{1}{2} m\omega^2 \left(\frac{(x_1^2 - a^2)^2}{4a^2} + y_1^2 \right) + \frac{1}{2} m\omega^2 \left(\frac{(x_2^2 - a^2)^2}{4a^2} + y_2^2 \right)$

$$- \frac{1}{2} m\omega^2 \left((x_1 - a)^2 + y_1^2 \right) - \frac{1}{2} m\omega^2 \left((x_2 - a)^2 + y_2^2 \right)$$

Note that $y_{1,2}$ cancels.

$$\varphi_{\pm a}(x,y) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar}(x \mp a)^2} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar} y_1^2}$$

is wave function of $n_x = 0, n_y = 0$ state of simple harmonic oscillator (SHO) centered at $(\pm a, 0)$

Heitler-London approximation

$$E_t - E_s = \langle \psi_- | V_{12} | \psi_- \rangle - \langle \psi_+ | V_{12} | \psi_+ \rangle$$

Observe that $V_{12}(x_1, x_2) = U(x_1) + U(x_2)$

where $U(x) = \frac{1}{2} m\omega^2 \left[\frac{(x^2-a^2)^2}{4a^2} - (x-a)^2 \right] =$

$$= \frac{1}{2} m\omega^2 (x-a)^2 \left[\frac{(x+a)^2}{4a^2} - 1 \right] = \frac{m\omega^2}{8a^2} (x-a)^2 \left[x^2 + 2xa - 3a^2 \right]$$

Thus $E_t - E_s = \frac{1}{2(1-s^2)} \int dx_1 dx_2 \, V_{12} \left(\varphi_a(x_1)\varphi_{-a}(x_2) - \varphi_a(x_2)\varphi_{-a}(x_1) \right)^2$

$$- \frac{1}{2(1+s^2)} \int dx_1 dx_2 \, V_{12} \left(\varphi_a(x_1)\varphi_{-a}(x_2) + \varphi_a(x_2)\varphi_{-a}(x_1) \right)^2 =$$

$$= \frac{1}{2(1-s^2)} \int dx_1 dx_2 \left(U(x_1) + U(x_2) \right) \Big[\varphi_a^2(x_1)\varphi_{-a}^2(x_2) + \varphi_a^2(x_2)\varphi_{-a}^2(x_1)$$

$$- 2\, \varphi_a(x_1)\varphi_{-a}(x_1)\, \varphi_a(x_2)\varphi_a(x_2) \Big]$$

$$- \frac{1}{2(1+s^2)} \int dx_1 dx_2 \left(U(x_1) + U(x_2) \right) \Big[\varphi_a^2(x_1)\varphi_{-a}^2(x_2) + \varphi_a^2(x_2)\varphi_{-a}^2(x_1)$$

$$+ 2\, \varphi_a(x_1)\varphi_{-a}(x_1)\, \varphi_{-a}(x_2)\varphi_a(x_2) \Big]$$

All integrals are symmetric under $1 \Leftrightarrow 2$ exchange of indices $\Rightarrow$ can keep only $U(x_1)$ under the integral and double the result:

$$E_t - E_s = \frac{1}{2(1-s^2)} \Big\{ 2 \int dx_1 \, U(x_1) \varphi_a^2(x_1) \underbrace{\int dx_2 \, \varphi_{-a}^2(x_2)}_{1} \cdot 2$$

$$- 2 \cdot 2 s \int dx_1 \, U(x_1) \varphi_a(x_1) \varphi_{-a}(x_1) \Big\} -$$

overlap $\hookrightarrow \int dx_2 \, \varphi_a(x_2)\varphi_{-a}(x_2)$

$$- \frac{1}{2(1+s^2)} \Big\{ 4 \int dx_1 \, U(x_1) \varphi_a^2(x_1) \underbrace{\int dx_2 \, \varphi_{-a}^2(x_2)}_{1}$$

$$+ 4 s \int dx_1 \, U(x_1) \varphi_a(x_1) \varphi_{-a}(x_1) \Big\} .$$

$$\Rightarrow E_t - E_s = \frac{4s^2}{1-s^4}\left\{\underbrace{\int dx\, U(x)\,\varphi_a^2(x)}_{\text{"Coulomb"}} - \underbrace{\frac{1}{s}\int dx\, U(x)\,\varphi_a(x)\varphi_a(x)}_{\text{"exchange"}}\right\}$$

$$I_1 = \int_{-\infty}^{\infty} dx\, U(x)\,\varphi_a^2(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\int dx\; e^{-\frac{m\omega}{\hbar}(x-a)^2}\,\frac{m\omega^2}{8a^2}(x-a)^2\left[x^2+2xa-3a^2\right]$$

let $z = x-a \Rightarrow x = z+a$

$$x^2+2xa-3a^2 = (z+a)^2+2a(z+a)-3a^2 =$$
$$= z^2+2az+a^2+2az+2a^2-3a^2 =$$
$$= z^2+4az$$

$$I_1 = \frac{m\omega^2}{8a^2}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\int_{-\infty}^{\infty} dz\; z^2\underbrace{(z^2+\cancel{4az})}_{\text{odd!}}\, e^{-\frac{m\omega}{\hbar}z^2} =$$

$$= \left(t = \sqrt{\frac{m\omega}{\hbar}}\,z\right) = \frac{m\omega^2}{8a^2}\,\frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} dt\,\left(\frac{\hbar}{m\omega}\right)^2 t^4\, e^{-t^2} =$$

$$= \frac{\hbar^2}{8ma^2}\,\frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} dt\; t^4 e^{-t^2}$$

$$I_2 = \int dx\;\frac{m\omega^2}{8a^2}(x-a)^2\left[x^2+2xa-3a^2\right]\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{m\omega}{2\hbar}\left[(x-a)^2+(x+a)^2\right]} =$$

$$= \frac{m\omega^2}{8a^2}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\int_{-\infty}^{\infty} dx\;\underbrace{(x^2-2xa+a^2)(x^2+2xa-3a^2)}\; e^{-\frac{m\omega}{\hbar}(x^2+a^2)}$$

$$x^4+\cancel{2x^3a}-3a^2x^2-\cancel{2x^3a}-4a^2x^2+\cancel{6a^3x}$$
$$+a^2x^2+\cancel{2a^3x}-3a^4 = x^4-6a^2x^2-3a^4$$

keep only even in $\underline{x}$ terms here!

let again $t = \sqrt{\frac{m\omega}{\hbar}}\, x$

$$I_2 = \frac{m\omega^2}{8a^2\sqrt{\pi}} \int_{-\infty}^{\infty} dt \left(\frac{\hbar^2}{m^2\omega^2} t^4 - 6a^2 \frac{\hbar}{m\omega} t^2 - 3a^4 \right) e^{-t^2} e^{-\frac{m\omega a^2}{\hbar}}$$

$$= \frac{\hbar^2}{8ma^2\sqrt{\pi}} e^{-\frac{m\omega a^2}{\hbar}} \int_{-\infty}^{\infty} dt \left(t^4 - 6a^2 \frac{m\omega}{\hbar} t^2 - 3a^4 \frac{m^2\omega^2}{\hbar^2} \right) e^{-t^2}$$

Note that $S = \langle \varphi_a(x) | \varphi_{-a}(x) \rangle = \int dx \left(\frac{m\omega}{\pi\hbar} \right)^{1/2} e^{-\frac{m\omega}{2\hbar}[(x-a)^2 + (x+a)^2]}$

$$= \left(\frac{m\omega}{\pi\hbar} \right)^{1/2} \int_{-\infty}^{\infty} dx\, e^{-\frac{m\omega}{\hbar}(x^2+a^2)} = \frac{1}{\sqrt{\pi}} \underbrace{\int_{-\infty}^{\infty} dt\, e^{-t^2}}_{\sqrt{\pi}} e^{-\frac{m\omega a^2}{\hbar}} = e^{-\frac{m\omega a^2}{\hbar}}$$

Finally, $\int_{-\infty}^{\infty} dt\, t^4 e^{-t^2} = \frac{3\sqrt{\pi}}{4}$

$$\int_{-\infty}^{\infty} dt\, t^2 e^{-t^2} = \frac{\sqrt{\pi}}{2}, \quad \int_{-\infty}^{\infty} dt\, e^{-t^2} = \sqrt{\pi}$$

$$I_2 = \frac{\hbar^2}{8ma^2} S \left(\frac{3}{4} - 6a^2 \frac{m\omega}{\hbar} \frac{1}{2} - 3a^4 \frac{m^2\omega^2}{\hbar^2} \right)$$

$$I_1 = \frac{\hbar^2}{8ma^2} \cdot \frac{3}{4}$$

$$E_t - E_s = \frac{4S^2}{1-S^4} \left(I_1 - \frac{I_2}{S} \right) = \frac{4S^2}{1-S^4} \frac{\hbar^2}{8ma^2} \left(\frac{3m^2\omega^2 a^4}{\hbar^2} + \right.$$

$$\left. + \frac{3m\omega a^2}{\hbar} \right) = \frac{S^2}{1-S^4} \frac{3}{2} \left(m\omega^2 a^2 + \hbar\omega \right)$$

For large distance $R = 2a$ between the dots, where Heitler-London approximation makes sense, $\underbrace{E_t - E_s}_{\text{exchange splitting } J} \approx \frac{3}{8} m\omega^2 R^2 e^{-\frac{m\omega}{2\hbar} R^2}$

Thus $J \sim e^{-m\omega R^2/2\hbar}$ is indeed exponentially small!